Haoyi WANG

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EDUCATION

DARTMOUTH COLLEGE (THAYER) Aug 2025 –
Ph.D. in Electrical Engineering *Expected June 2027*

UNIVERSITY OF CONNECTICUT (UCONN) Aug 2022 – Aug 2025
Ph.D. in Electrical Engineering GPA: 3.9 / 4.0

SHANGHAI JIAO TONG UNIVERSITY (SJTU) Sep 2019 – Mar 2022
M.S. in Electrical Engineering GPA: 3.61 / 4.0

NORTH CHINA ELECTRIC POWER UNIVERSITY (NCEPU) Sep 2015 – Jun 2019
B.E. in Electrical Engineering and Automation GPA: 3.68 / 4.0

SKILLS

Programming and tools: Python, MATLAB, C, OpenDSS, Gurobi, CYME, RSCAD

Machine learning: TensorFlow, scikit-learn (GNNs, SVMs, ANNs)

Power systems: Distribution modeling, robust estimation, DER optimization, Volt-Var control

Electrical fundamentals: Personally instructed students in using **digital multimeters and oscilloscopes** for circuit measurements, testing, and troubleshooting.

RESEARCH & WORK EXPERIENCES

Project 1: BAME (Bus-Admittance Matrix Estimation) June 2026 – Sep 2026
ABB U.S. Corporate Research Center *ABB-USCRC*

- Developed a physics-constrained GAT framework for a 37-node distribution feeder, reducing 666 possible bus pairs to 94–95 candidates and identifying radial connectivity across 6 topologies.
- Formulated an unbalanced IEEE 34-bus parameter-estimation problem to correct utility line R/X matrices and the resulting Y-bus, estimating 304 independent parameters across 32 lines with SCADA measurement uncertainties.
- Proposed an uncertainty-aware two-stage LinDistFlow-to-AC WLS/MAP estimator with covariance-informed state estimates and line-code priors spanning 50%/80% initial error; LDF initialization reduced normalized RMSE by 53.2%/49.9% versus pure-global initialization and improved the original utility parameters by 54.1%/61.3%.
- Achieved final MADR of 9.16%/12.36% for enabling practical SCADA/EMS grid-model correction from limited, noisy measurements, the related work has been approved by the business unit for a patent submission.

Project 2: Data-Driven LSF-Based DER Dispatch Scheme in Distribution Systems Sep 2022 – May 2025
Research Assistant (*Individually investigated****)*** *Electrical and Computer Engineering Department, University of Connecticut*

- Constructed a novel measurement feedback online DER dispatch methodology and derived a linear programming (LP) formulation to improve traditional system optimization process and achieved real-time coordination. (Published papers)
- Proposed a robust recursive sparse voltage sensitivity estimation method to fast update sensitivity factors during system operation in less than 30 seconds computation.
- Formulated a scenario-based chance-constrained optimization to endure over 25% voltage and power measurement uncertainties and guaranteed an accurate estimation result with less than 3% error.
- Mitigated the overvoltage issues caused by over 50% PV penetration and a rapid online management system with 0.012 second computation per control, the total power curtailment is only 1.82% while traditional Volt-Var control is 4.07%.

Project 3: Local Autonomous Volt-Var Optimization in Distribution Grid (NLR Internship) May – Aug 2024

Graduate Researcher *(Individually reported in weekly meeting)* *NLR & Eaton Corporation*

- Investigated the feasibility and benefits of a localized Volt-Var optimization control strategy with 75% PV penetration, in alignment with EATON's grid-edge intelligence objectives.
- Designed a machine learning-based DER dispatch scheme to satisfy the practical deployment while eliminating the requirements for real-time data communications, which enhanced the response speed of system operators.
- Trained and tuned SVMs to classify optimal transformer tap settings (98% accuracy) and ANNs to compute optimal PV inverter setpoints (96% accuracy) for mitigating voltage violations. (Published paper)
- Accomplished all models on offline historical/simulated datasets and validated the complete dispatch scheme on a real Colorado distribution system (3384 nodes), demonstrating strong control performance and implementation readiness.

Project 4: Northeast University Cybersecurity Center for Advanced and Resilient Energy Delivery Oct 2024 –

Research Assistant (*Collaboration with eight institutes****)*** *Thayer School of Engineering, Dartmouth College*

- Reviewed, collected and documented over 10 different attack spaces with relevant research work and industrial protocols (i.e, IEEE standard, NIST, SunSpec, IEC) for the corresponding features and mitigation measures.
- Designed and implemented a grid-aware optimization framework to mitigate cyberattack effects by coordinating distributed energy resource (DER) using local measurements and communications, the computing burden is second level that can outperform while system operation.
- Developed and prototyped an admittance-matrix-based sensitivity adjustment method to accommodate power curtailment under topology changes to less than 3% (used to be 15%) and reduced the overvoltage nodes to only 1% in a real system.
- Proposed a topology reconfiguration mechanism to mitigate the voltage violation to less than 5% (used to be 30%) caused by PV inverter shut off and thus improved distribution system resilience. (TSG second round review)

INNOVATIVE ACTIVITIES

Solar Forecasting Behind the Meter PV Generation Forecasting System Jun 2023 – Aug 2023

Advisors: *Junbo Zhao, Zhu Zhe* ***Eversource*** *& University of Connecticut*

- Mapped PV installations using **0.6-m NAIP aerial imagery and 3-m PlanetScope satellite imagery**; forecasted feeder-level PV injections using irradiance, panel orientation, temperature, and network constraints.
- Collaborated with a civil engineering Ph.D. student and mentored an electrical engineering undergraduate; won **second prize ($2,500 per team member)**.

AWARDS

- **BEST POSTER AWARD** in the ECE Department in the University of Connecticut (**3** out of **55 Ph.D.** students) 2023
- **EVERSOURCE CLEAN ENERGY & SUSTAINABILITY FELLOWSHIP –** UConn (**18** out of **200**) 2023

PUBLICATIONS

W. Haoyi, L.Yingqi, Y. Yiyun, Z. Junbo and D. Fei, "**Online Model-Free DER Dispatch via Adaptive Voltage Sensitivity Estimation and Chance Constrained Programming**," in IEEE Transactions on Power Systems, no. 6, pp. 7318-7330, 2024.

W. Haoyi, Y. Yiyun, Z. Junbo and D. Fei, "**Data-Driven Mean-Corrected Recursive Estimation-based Optimal DER Dispatch for Distribution System Voltage Control**" in IEEE Transactions on Sustainable Energy, 2025.

More publications on google scholar page: https://scholar.google.com/citations?user=8MbjCIUAAAAJ&hl=en